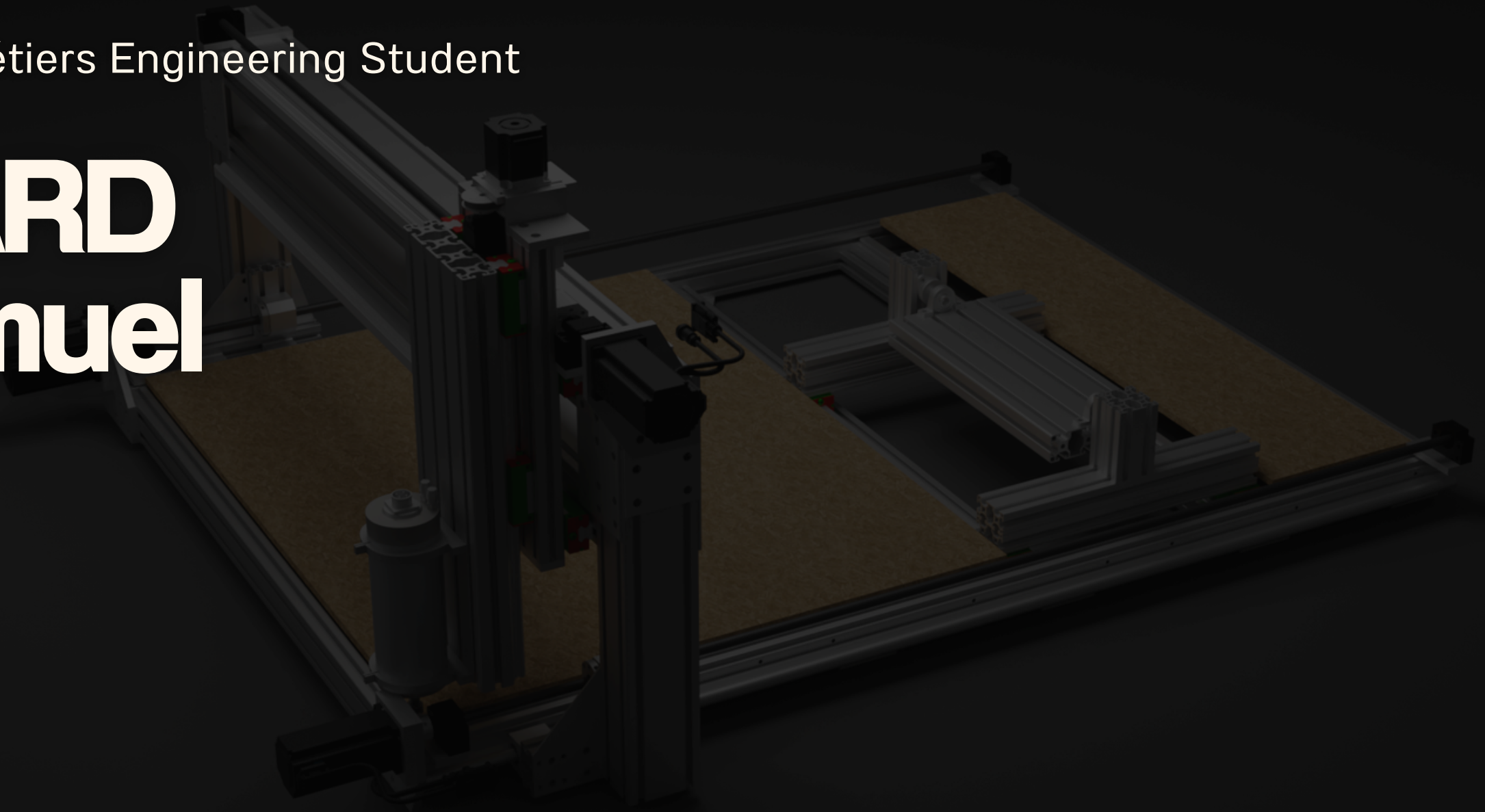


Arts et Métiers Engineering Student

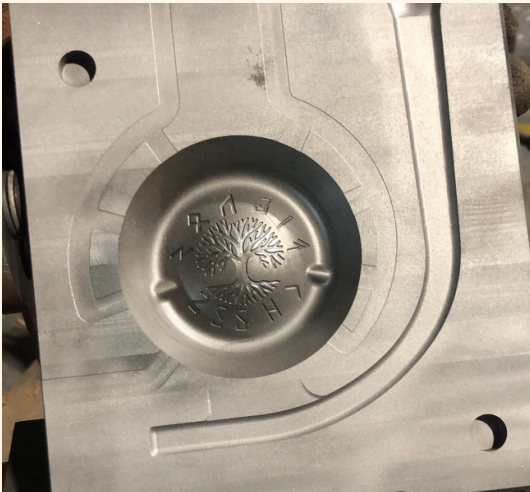
DIARD Samuel

PORTFOLIO

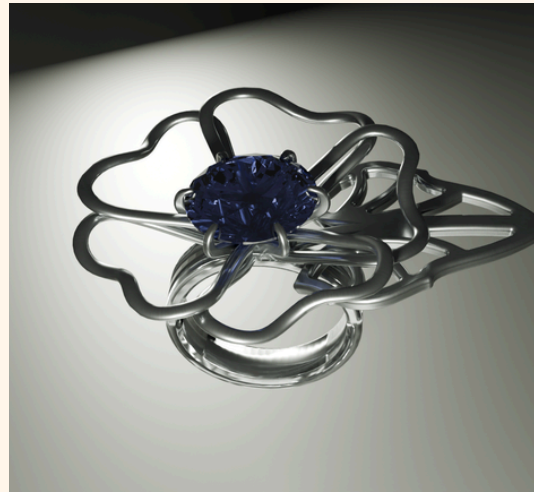


Projects

**INDUSTRIALISATION -
METAL MOULDS & SERIAL
PRODUCTION**



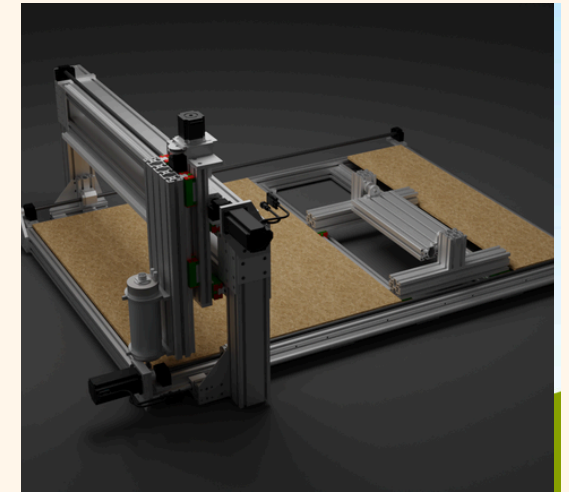
**JEWELLERY - LOST-WAX
CASTING**



**THERMAL ENGINEERING -
1200°C PROGRAMMABLE
FURNACE**



**PRECISION ENGINEERING -
5-AXIS CNC MILLING
MACHINE**



Industrialisation – Metal Moulds & Serial Production

CONTEXT:

For over half a century, producing a commemorative ashtray has been a tradition of the Arts et Métiers Grand Gala des Fignos. For the 2024 edition, a new model was designed around the event's theme (The Vikings), with the goal of continuing this tradition through the production of 300 aluminium pieces. I was responsible for the project, developing the entire manufacturing process, from ashtray design to mould production and final casting.

OBJECTIVES :

Design and implement a manufacturing process for the serial production of these ashtrays, from part design to final production.

The project included:

- ashtray design;
- development of the casting system;
- design and machining of cast-iron moulds;
- aluminium casting.



One of the two mould halves



An ashtray after casting and finishing

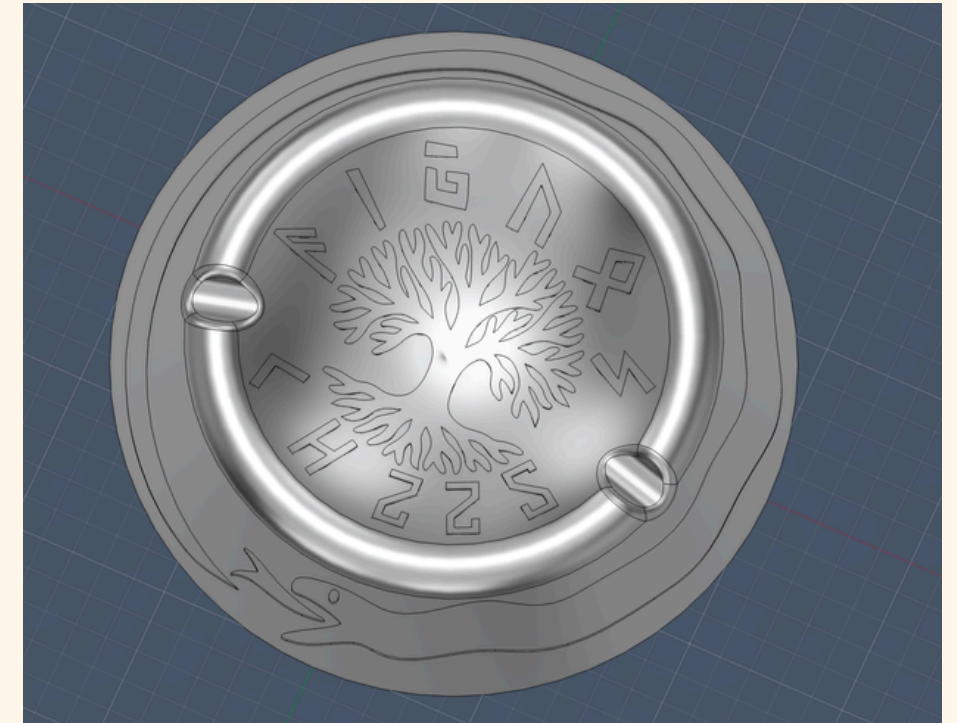
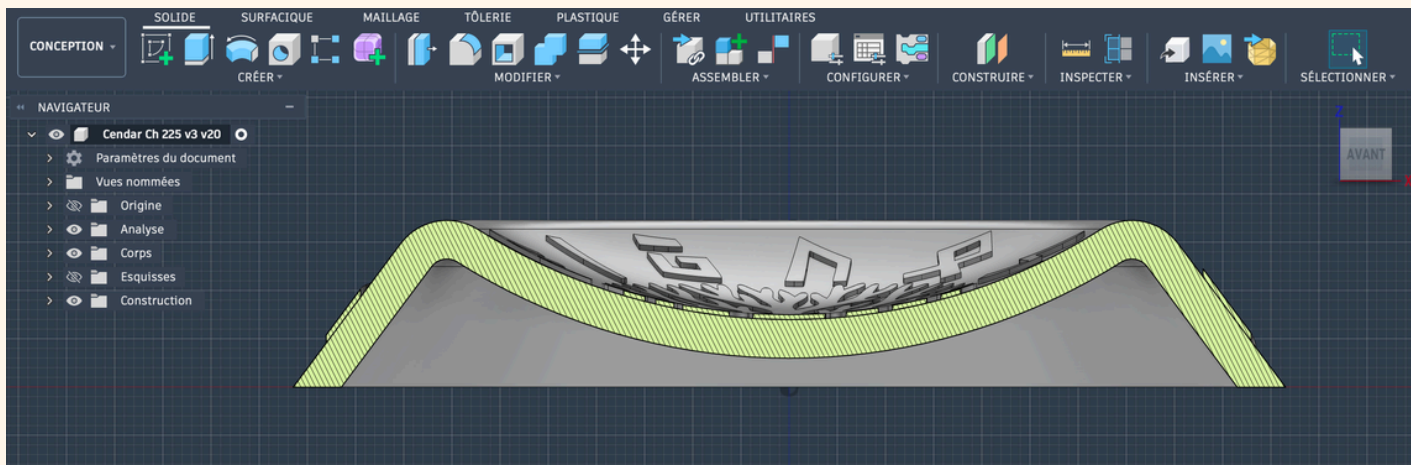
Industrialisation – Metal Moulds & Serial Production

ASHTRAY DESIGN:

CAD design integrating permanent mould casting constraints from the outset: no undercuts, draft angles for demoulding, and uniform wall thickness to prevent shrinkage defects.

Particular attention was paid to demoulding: undercuts were avoided, vertical walls were angled to ease extraction, and uniform wall thickness was favoured to limit shrinkage during solidification.

Patterns and inscriptions were integrated directly into the model with additional material, creating recessed features in the mould and simplifying machining.



3D model of the ashtray in sectional and top views. CAD designed in Fusion 360.

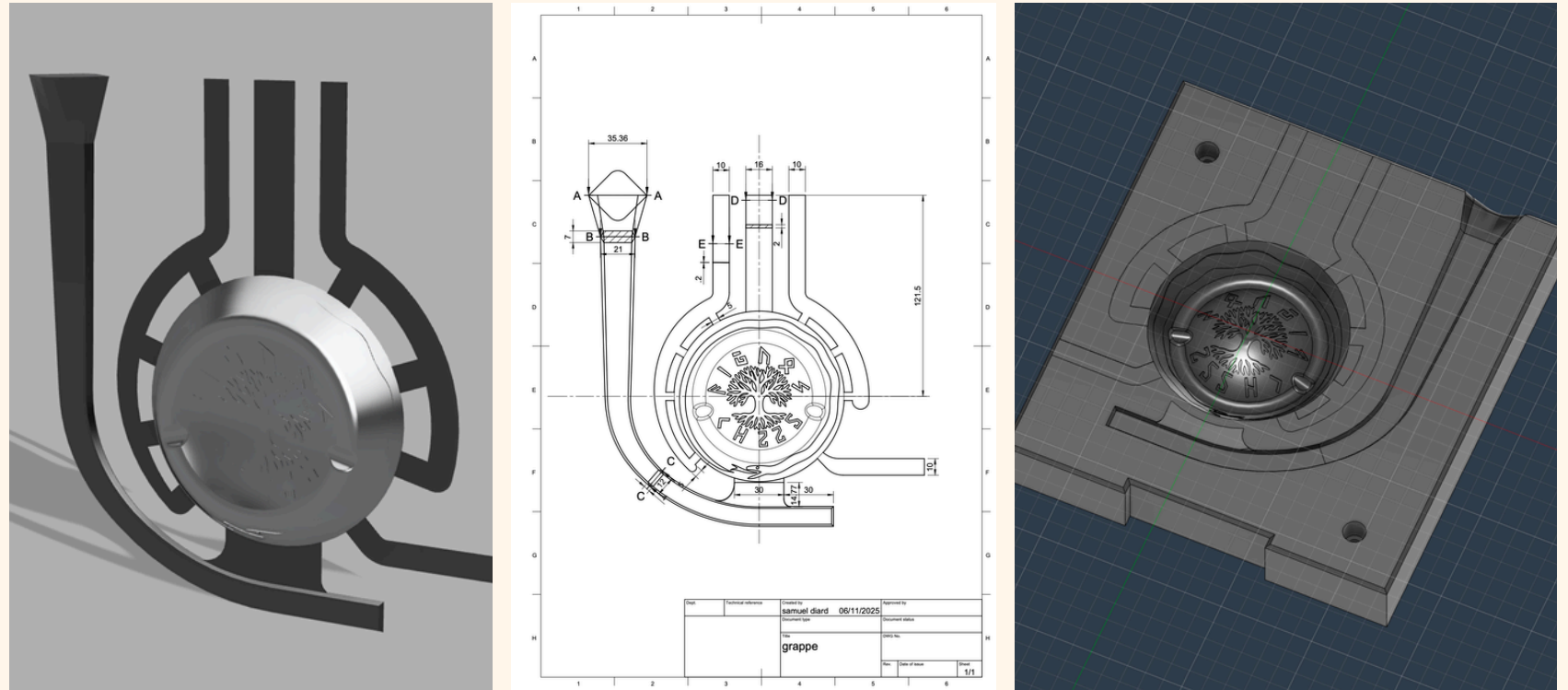
Industrialisation – Metal Moulds & Serial Production

MOULD DESIGN:

Two-part permanent cast-iron mould, with the casting system sized directly in CAD for proper filling without rework across all 300 parts.

A two-part permanent cast-iron mould was designed to facilitate opening and part removal after solidification. The parting planes, support surfaces and guiding features were designed to ensure accurate positioning and good repeatability throughout the casting cycles.

The casting system (runners, gates and vents) was sized using technical literature and guidance from the foundry professor, ensuring good castability from the first trials and limiting defects. This approach avoided lengthy and complex modifications to the mould later on.



CAD of the casting system, its technical drawing, and one of the two mould halves

Industrialisation – Metal Moulds & Serial Production

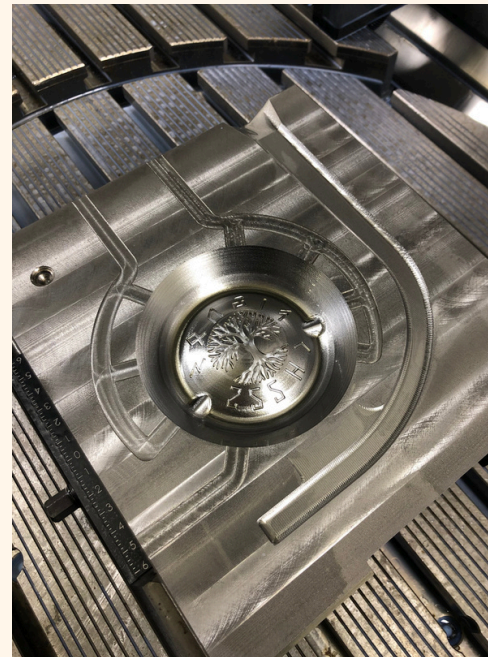
MOULD MANUFACTURING:

EN-GJL-250 cast-iron blocks ground and CNC-machined from the CAD model. Tapping, sandblasting and mould coating completed the tooling before the first casting, ensuring accuracy and clean demoulding.

The two mould halves were made from EN-GJL-250 grey cast-iron blocks (250 × 250 × 40 mm). The support surfaces were first ground on a conventional milling machine to obtain precise reference geometry, before machining the cavities on a CNC milling machine from the CAD model.

Additional operations added the features required for mould use: M14 tapped holes for handles and clamps, preparation of functional surfaces, and adjustment of assembly elements.

Before the first casting, the moulds were sandblasted and baked to remove cutting oil residues absorbed by the cast iron during machining. A mould coating was then applied to the surfaces in contact with aluminium, protecting the mould, facilitating demoulding and improving the surface quality of the parts.



Mould halves after machining and completed production tooling

Industrialisation – Metal Moulds & Serial Production

PART PRODUCTION:

300 ashtrays cast at 700°C in moulds preheated to 350°C, at a rate of around one part per minute. The casting system was designed for removal without sawing, reducing finishing time.

The moulds were preheated to around 350°C to limit thermal shock, promote uniform cavity filling and improve part quality. Aluminium was then cast at 700°C directly into the mould, enabling a production rate of around one part per minute. The mould was also designed to optimise downstream operations: the gates were positioned and sized to allow easy removal of the casting system after solidification, without sawing. The runner could simply be broken off mechanically, reducing finishing time.



Removal of casting gates and final result after deburring

Industrialisation – Metal Moulds & Serial Production

CONCLUSION :

This project resulted in a complete series of 300 aluminium ashtrays through a fully controlled manufacturing process, from part design to finished production. The tooling and casting system were designed to ensure repeatable production from the first trials while limiting finishing operations.

This project developed my overall approach to product development by integrating manufacturing constraints from the design stage. It particularly taught me the importance of anticipating production steps to design not only a functional part, but also a reliable and repeatable process.

KEY FIGURES:

- 300 parts produced
- Over 40 h of CNC and conventional machining
- 50 kg of aluminium
- 35 kg of cast iron
- 60 s cycle time

From digital design to final part, this project reflects my approach of developing solutions where technical expertise and process knowledge serve a concrete result.

SKILLS DEVELOPED:

Mechanical Design:

- Designing a part for a casting process
- Designing production tooling
- Integrating manufacturing constraints from the CAD stage

Manufacturing Processes:

- CNC machining of cast-iron moulds
- Aluminium permanent mould casting
- Developing and optimising a production process

Project Methodology:

- Managing a complete development process
- Solving technical challenges
- Moving from prototype to small-scale production

Jewellery – Lost-Wax Casting

CONTEXT:

This ring was made for my parents' 25th wedding anniversary. They provided the stone, a 7.79-carat lab-grown blue diamond, with the intention of having it set in a unique piece entirely designed and crafted by me. The

OBJECTIVES :

The design was developed around my mother's requirements, integrating the constraints of lost-wax casting and stone setting from the first sketches – model castability and mechanical strength of the setting around a 7.79-carat stone.

The project covered the entire manufacturing process: CAD design, casting system development, wax model printing, silver lost-wax casting, polishing and final stone setting.



Finished ring and 3D rendering

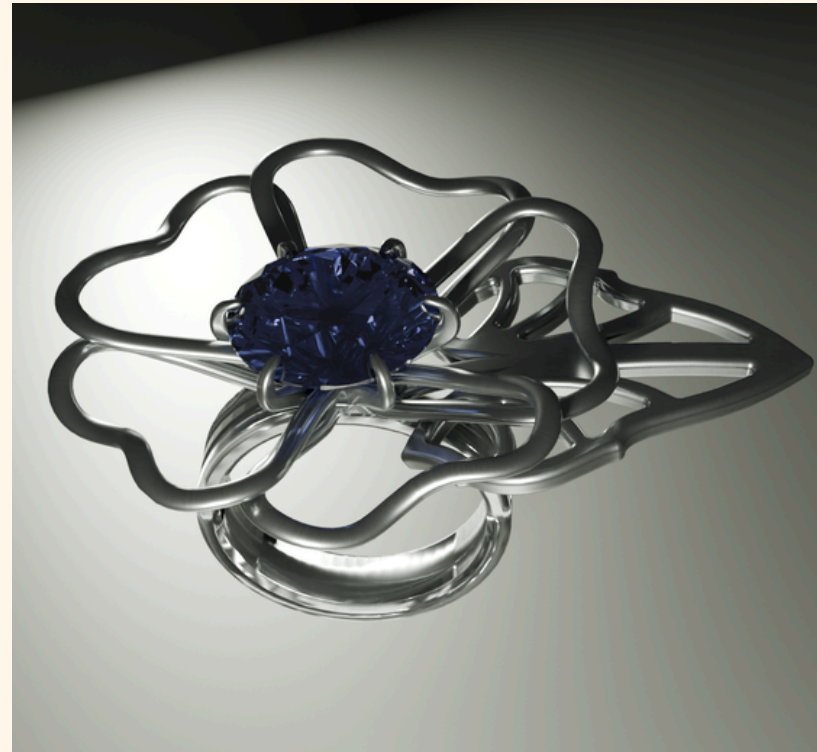
Jewellery – Lost-Wax Casting

CAD & 3D PRINTING:

3D model created using Fusion 360 and Blender, before integrating the casting system. The ring was then printed in castable resin, ready for casting.

Once the ring shape was defined, the 3D model was developed using both Fusion 360 and Blender: the former for geometric accuracy and mechanical constraints, the latter to refine the flower's organic forms. The casting system (sprue and positioning) was then integrated directly into the model.

The ring was finally printed in castable resin (PAC10 HeyGears) using a 3D printer, producing a “wax” model directly suitable for casting.



Blender rendering of the CAD model and wax model after 3D printing

Jewellery – Lost-Wax Casting

CASTING:

The wax model is embedded in a casting flask filled with degassed investment plaster, then fired for around 12 h up to 730°C to remove the wax. Molten silver (1000°C) is cast under vacuum at 550°C, followed by plaster removal.

The wax model is placed in a steel casting flask, which is then filled with investment plaster. The plaster is degassed before and after pouring to remove air bubbles that could cause defects in the final part. The flask is then fired for around twelve hours, reaching a maximum temperature of 730°C to fully remove the wax and cure the plaster. It is cooled to 550°C before molten 925 silver is poured at 1000°C under vacuum to ensure complete filling of the mould.

Once the part has solidified, the plaster is broken away to release the casting.



Investment plaster in the flask, metal casting, and raw casting with the stone on top

Jewellery – Lost-Wax Casting

POLISHING & STONE SETTING:

Acid cleaning, removal of casting gates, then progressive polishing to a mirror finish (sandpaper, brushes, felt and red/black compounds, with ultrasonic cleaning between each step). Prong setting, with prong length anticipated during casting, requiring only bending over the stone.

The casting is first acid-cleaned to remove oxides formed during casting, then the gates are removed with pliers and a saw. Coarse sanding removes the main surface imperfections, followed by cleaning with a Dremel-mounted brush.

The piece is then polished with felt and polishing compound – first red, then black for a mirror finish – with ultrasonic cleaning between each stage to remove compound residue and visually monitor the progress.

The stone is finally secured using prong setting, with the prong length anticipated during the design and casting of the setting. Each prong is then bent over the stone, taking care not to apply excessive force given its size.



Ring during polishing and final result after stone setting

Jewellery – Lost-Wax Casting

CONCLUSION :

This project allowed me to take a unique piece of jewellery from start to finish, from 3D design to final stone setting, integrating the constraints of lost-wax casting and the setting of a large stone from the design stage. Beyond the technical aspect, this ring combines manufacturing precision with aesthetic requirements – two key principles in my approach to jewellery making. This project also allowed me to work for a client with precise expectations, in this case my mother, and deliver a piece meeting both strong aesthetic requirements and demanding technical constraints.

KEY FIGURES:

- 7.79 carats (Fancy Vivid Blue diamond)
- 13 g (925 silver)

Creating this ring allowed me to develop a comprehensive approach to jewellery making, from digital design to manual finishing. This project taught me the importance of anticipating the manufacturing constraints of lost-wax casting and stone setting from the CAD stage, in order to create a piece that is both aesthetically refined and technically controlled.

SKILLS DEVELOPED:

Design

- Mixed CAD modelling (Fusion 360 / Blender)
- Design integrating casting and stone-setting constraints

Manufacturing Process

- Castable resin 3D printing
- Silver lost-wax casting

Jewellery

- Polishing
- Prong setting

Thermal Engineering – 1200°C Programmable Furnace

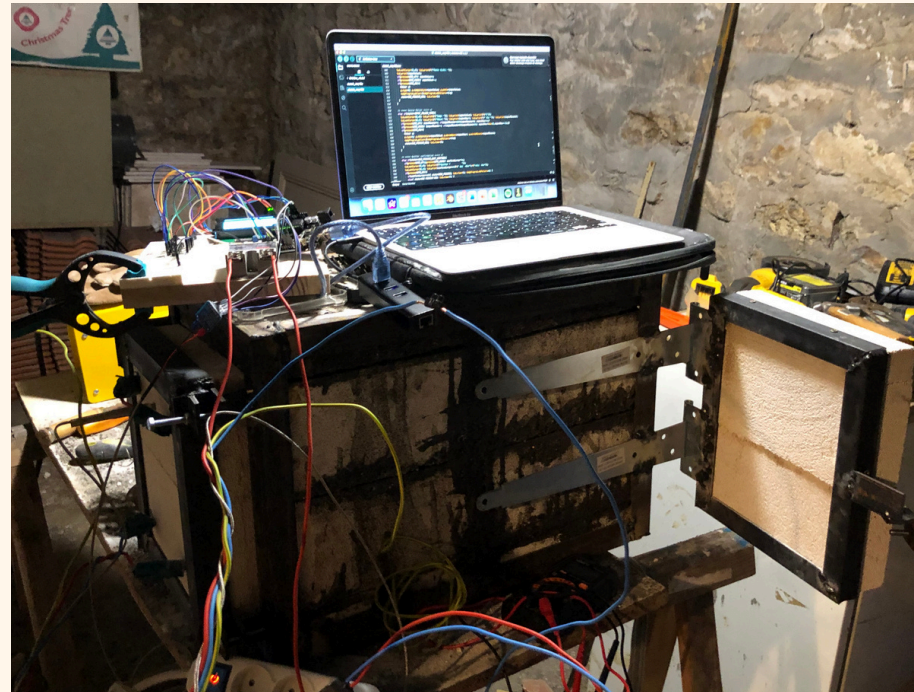
CONTEXT:

Forging and lost-wax casting require precise control of thermal cycles. As part of my personal projects, I developed a programmable electric furnace for two distinct applications: forging, particularly knife making, requiring precise temperature control, and firing the plaster moulds used in lost-wax casting, which requires an automatic temperature profile to gradually remove the wax without damaging the mould.

OBJECTIVES :

Design an electric furnace capable of reaching 1200°C with stable temperature control and versatile operation. The goal was to create a reliable, safe tool suited to different manufacturing processes through two operating modes.

- Hold mode: stable temperature for forging or heat treatment
- Program mode: automatic temperature profile for processes requiring precise control



Furnace and electronics under development, and knife forging

Thermal Engineering – 1200°C Programmable Furnace

ARCHITECTURE:

Welded steel frame, Class 26 refractory bricks and ceramic fibre insulation. Heating is provided by custom-made 3 kW Kanthal A1 elements, controlled by an Arduino via SSRs and a Type K thermocouple.

The furnace combines Class 26 refractory bricks, chosen for their resistance up to 1430°C and their balance between mechanical strength and thermal insulation. They are assembled with refractory mortar in a welded steel frame, then covered with ceramic fibre to reduce heat losses and improve energy efficiency.

Heat is generated by custom-made 3 kW Kanthal A1 heating elements, regulated by modulating a solid-state relay (SSR) controlled by an Arduino UNO, with a Type K thermocouple for precise temperature monitoring.



Machining the grooves for the heating elements and assembled furnace with its door

Thermal Engineering – 1200°C Programmable Furnace

HEATING ELEMENT MANUFACTURING:

Helical Kanthal A1 heating elements sized for 3 kW at 230 V (13 A, 18 Ω), custom-formed from resistance wire.

The goal was to design helical Kanthal A1 heating elements, a ferritic iron-chromium-aluminium alloy known for its mechanical strength and oxidation resistance at high temperatures, for a total power of 3 kW.

To reach 3 kW at 230 V, the heating element must carry approximately 13 A, corresponding to a target resistance of around 18 Ω . The required wire length was then determined from the resistivity of Kanthal A1 and the selected cross-section, balancing high-temperature mechanical strength, size and manufacturing cost.

The heating element was made from Kanthal wire wound using a wooden jig and an axis driven by a power drill.



Resistance wire, formed heating element and element in operation inside the furnace

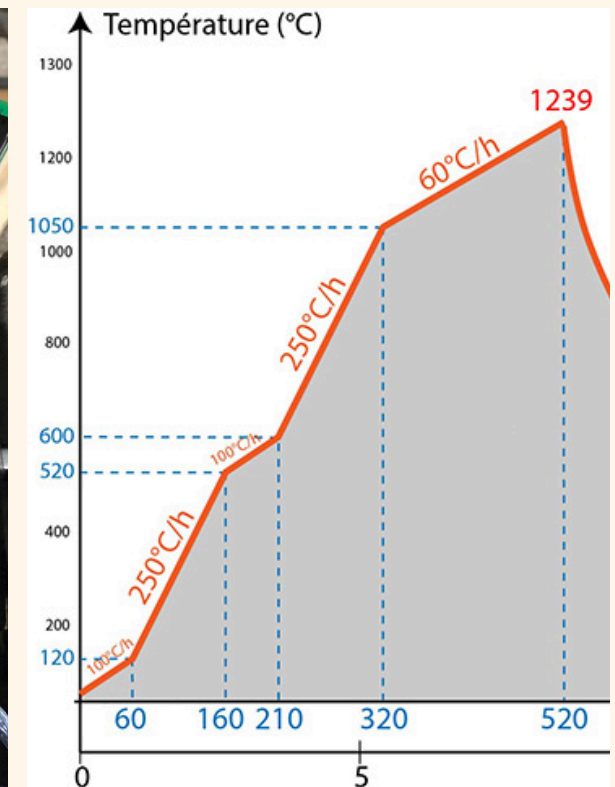
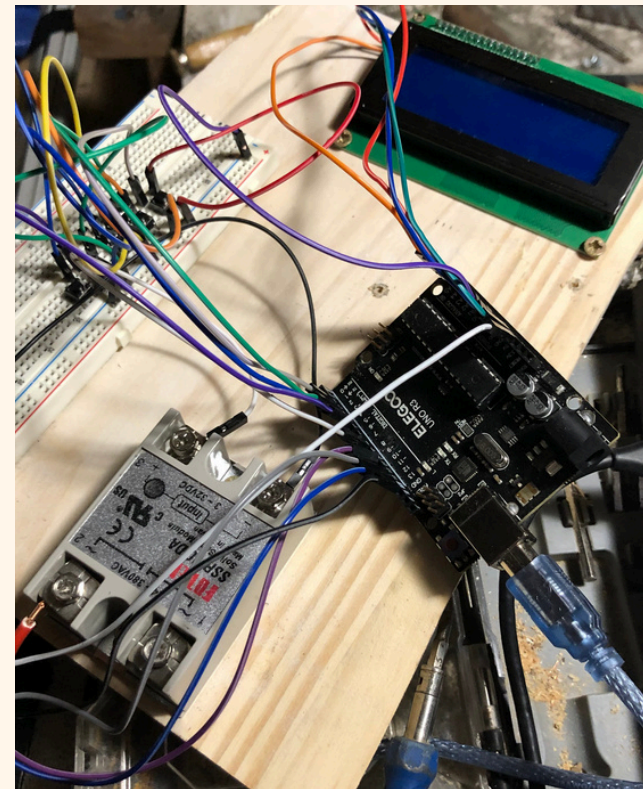
Thermal Engineering – 1200°C Programmable Furnace

ELECTRONICS:

Time-proportional control: an SSR driven by an Arduino UNO alternates between full power and off according to the setpoint. The furnace's thermal inertia smooths these variations for stable control, either at a fixed temperature or along a programmed profile.

To precisely control the temperature, two approaches were considered: continuously varying the voltage across the heating elements, which is complex at this power level, or using time-proportional control. The latter was chosen: the furnace alternates between full-power and unpowered phases, with the thermal inertia of the system naturally smoothing these rapid variations.

Switching is handled by a solid-state relay (SSR), chosen over an electromechanical contactor for its durability, lack of mechanical wear and ability to withstand a high number of cycles. The SSR is controlled by an Arduino UNO receiving data from a Type K thermocouple (measurements up to $\approx 1250^{\circ}\text{C}$), allowing the furnace to maintain a fixed setpoint or automatically follow pre-programmed firing cycles, particularly for lost-wax casting.



Assembled electronics and example of a firing profile

Thermal Engineering – 1200°C Programmable Furnace

CONCLUSION :

This project resulted in the design and construction of a programmable electric furnace capable of meeting various thermal processing needs. Its electronic control system allows it to maintain a set temperature or automatically follow complex firing cycles, particularly for preparing refractory plaster moulds used in lost-wax casting.

KEY FIGURES:

- 1200°C
- 3000 W
- 38 m of resistance wire
- Over 600 lines of code

Building this furnace allowed me to develop a comprehensive approach to industrial equipment design, integrating mechanical, thermal and electronic constraints. This project taught me the importance of component sizing and the interaction between different subsystems when designing a machine.

SKILLS DEVELOPED:

Mechanical Design

- Design of a welded steel structure
- Selection of high-temperature materials
- Design of a refractory assembly

Thermal Engineering & Processes

- Sizing of a heating system
- Thermal cycle management

Electronics & Control

- Power control using SSRs
- Temperature measurement using a thermocouple
- Arduino programming and temperature control

Precision Engineering – 5-Axis CNC Milling Machine

CONTEXT:

I had previously built a 3-axis CNC milling machine based on a design found online. This project allowed me to become familiar with the mechanical design of this type of machine, particularly its rigidity and accuracy requirements. The machine was eventually sold, which motivated me to take the project further.

Building on this experience, I decided to design a new machine entirely from scratch, without relying on an existing design, and add two additional axes to move from a 3-axis to a 5-axis architecture.

OBJECTIVES :

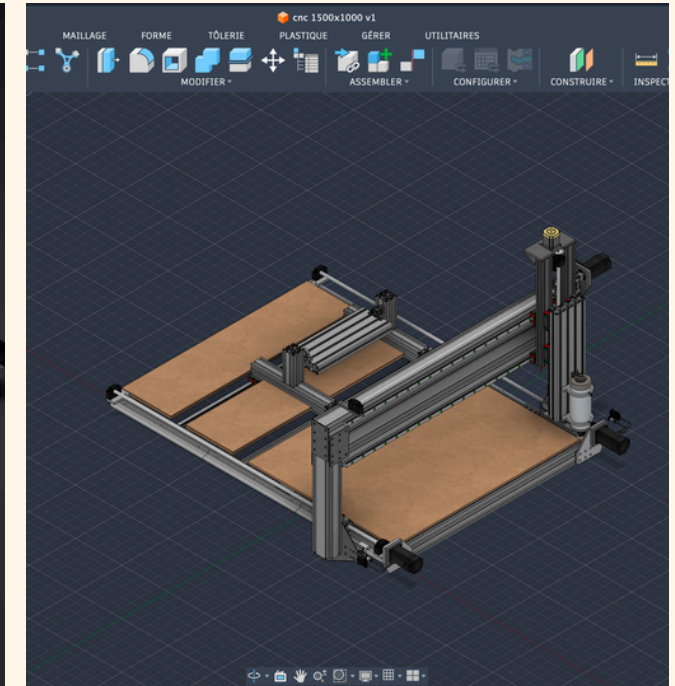
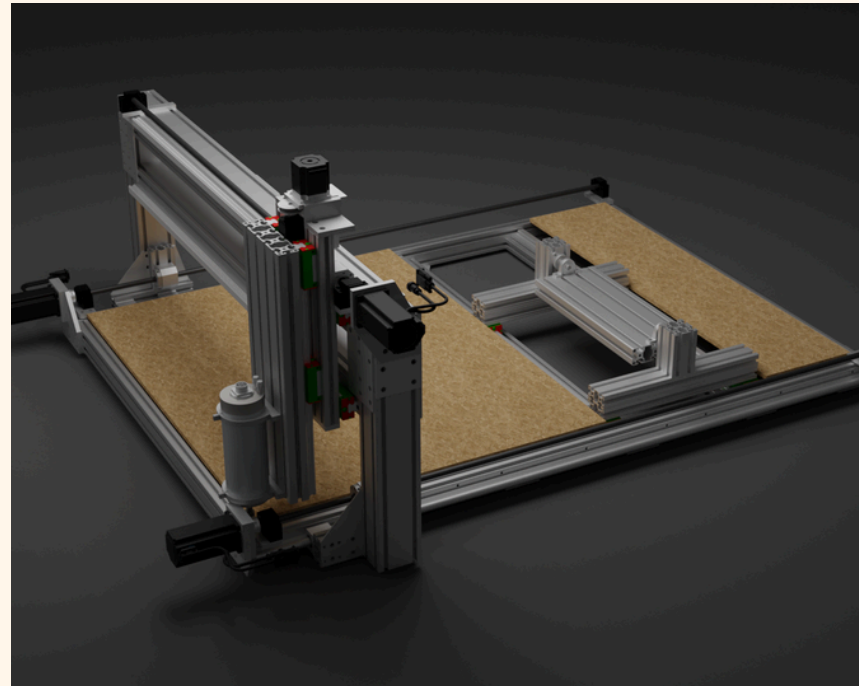
Design and build a CNC milling machine capable of multi-axis machining.

Design a mechanical architecture enabling simultaneous positioning of 5 axes.

Size and integrate the different subassemblies: structure, guides, ball screws, spindle and drive system.

Develop the control electronics and integrate the different axes.

Find a balance between rigidity, accuracy, footprint and cost.



3D Rendering and CAD

Precision Engineering – 5-Axis CNC Milling Machine

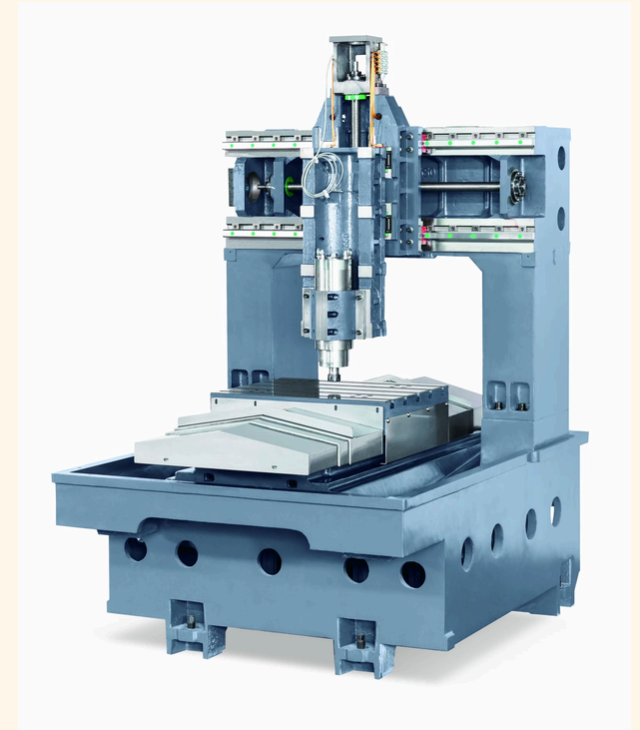
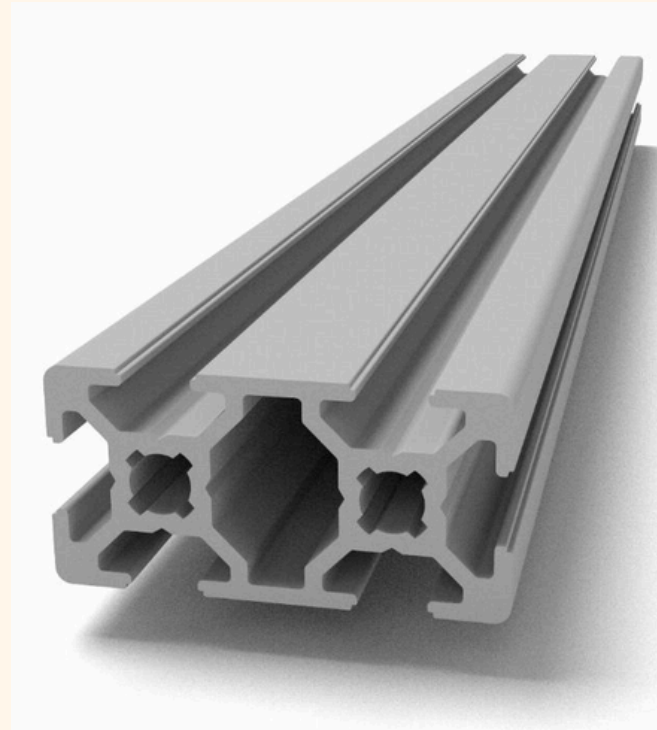
TECHNICAL SOLUTIONS:

Several structural solutions were considered: welded steel, epoxy granite, or bolted aluminium profiles. The latter was chosen as the best compromise between rigidity, ease of manufacturing, weight and the ability to build a large-volume machine.

The machine frame is made from bolted aluminium profiles. This solution allows rapid fabrication and relatively simple alignment, while providing sufficient rigidity for the targeted machining volume.

A welded steel frame would have provided greater rigidity, but welding distortion would have made guide alignment more complex. Conversely, an epoxy-granite frame would have offered excellent rigidity and vibration damping, but at the cost of more complex fabrication, higher mass and less suitability for a large-volume machine.

The choice of aluminium profiles therefore results from a compromise between rigidity, accuracy, ease of fabrication, mass and machining volume.



Aluminium Profile and Example of a Steel CNC Machine Frame

Precision Engineering – 5-Axis CNC Milling Machine

DESIGN:

The machine was designed in Fusion 360 using CAD models of components provided by manufacturers. The architecture is based on linear rails, ball screws and closed-loop NEMA 34 stepper motors, selected for their accuracy and reliability.

The machine was designed in Fusion 360, integrating the CAD models of the aluminium profiles provided by the supplier, allowing the design to be built directly around the actual component dimensions.

The axes are guided by linear rails, providing high accuracy and load capacity. Motion is transmitted by ball screws, selected for their accuracy and low friction, thereby limiting power losses.

The drive system uses closed-loop NEMA 34 stepper motors. They offer a compromise between the simplicity of conventional stepper motors and the higher performance, cost and complexity of servo motors.



NEMA 34 motor, its driver, HGR20 linear rails, and SFU1605 ball screws

Precision Engineering – 5-Axis CNC Milling Machine

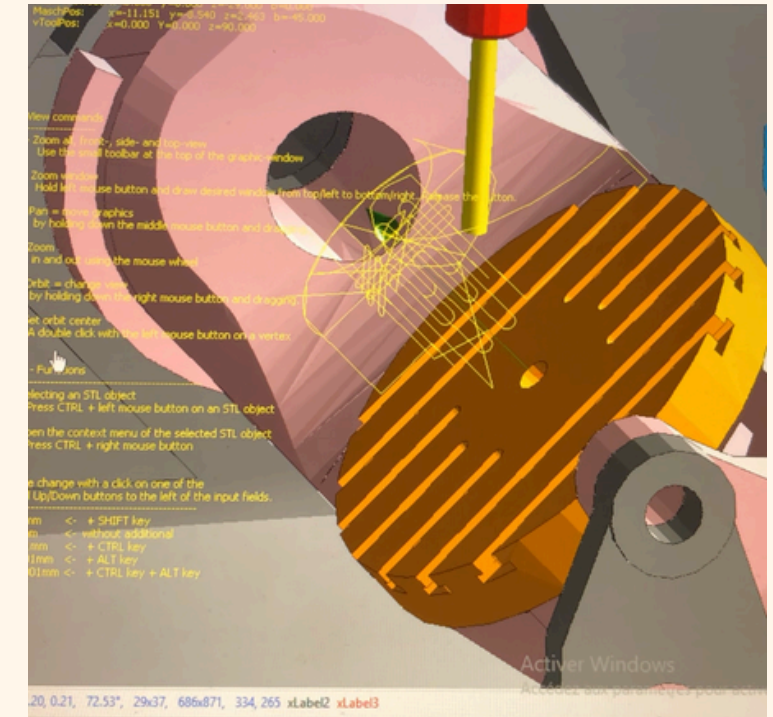
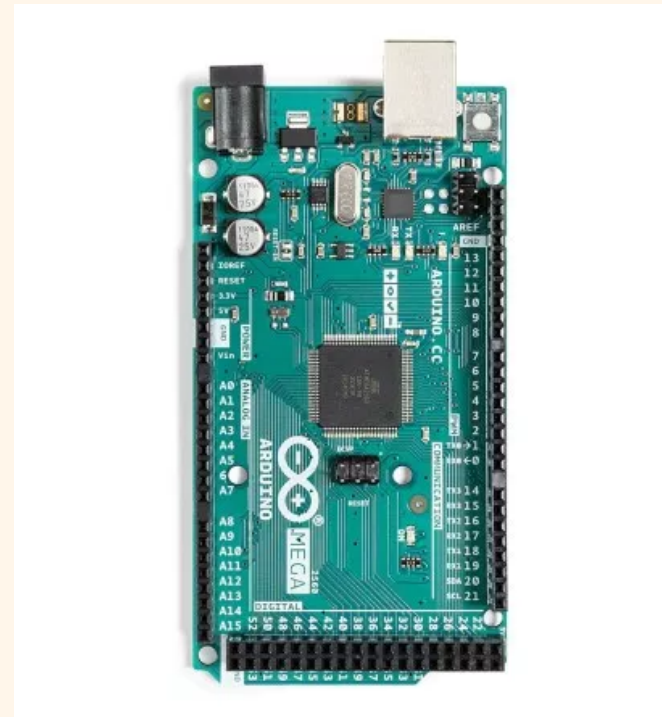
ELECTRICAL ARCHITECTURE:

The machine is controlled from a computer via GRBLGru, which communicates serially with an Arduino Mega equipped with a CNC Shield. The controller runs GRBL Mega 5X to manage the five axes, while dedicated drivers operate the closed-loop motors. A 2.2 kW liquid-cooled spindle is also controlled by a variable frequency drive (VFD).

The machine is controlled from a computer using the GRBLGru software, which transmits commands via a serial connection to an Arduino Mega equipped with a CNC Shield. The controller runs GRBL Mega 5X, an adaptation of GRBL that enables the management of five axes on an Arduino Mega.

The control signals are then transmitted to the motor drivers, which handle motor operation as well as the position feedback required for closed-loop control.

Machining is performed using a liquid-cooled spindle driven by a 2.2 kW variable frequency drive (VFD). This drive notably allows the spindle speed to be adjusted according to the machining requirements.



Arduino Mega controller board and machining simulation in GRBLGru

Precision Engineering – 5-Axis CNC Milling Machine

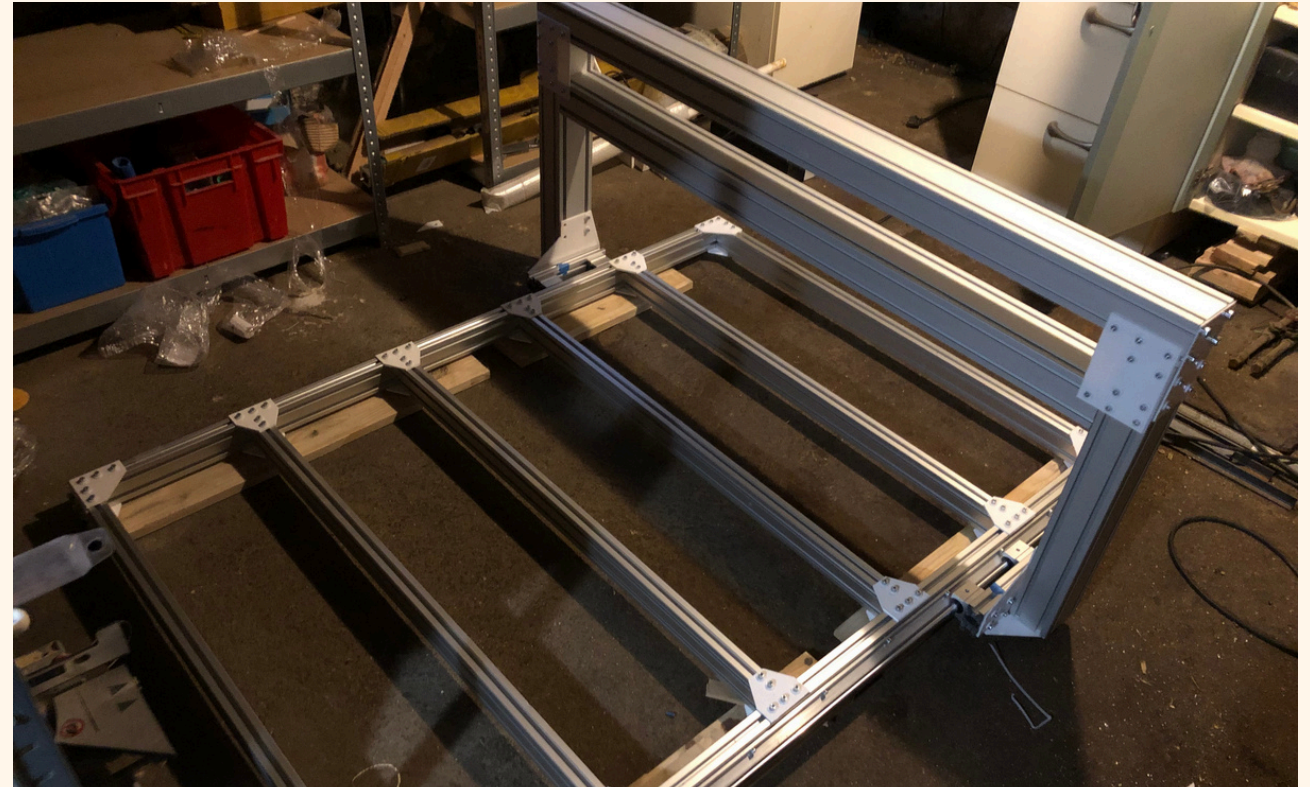
MANUFACTURING:

The construction of the machine is currently in progress. The fixed frame and the moving gantry forming the Y-axis have already been completed, while some parts are temporarily manufactured using 3D printing to enable assembly and initial testing.

The machine is currently under construction. The fixed frame and the moving gantry of the Y-axis have already been manufactured and assembled, allowing the gradual integration of the various components.

Some parts have been intentionally produced as temporary 3D-printed components. These parts make it possible to validate the assembly and operation of the machine before manufacturing the final versions in aluminum or steel.

Once the machine is sufficiently operational, the first final components will be machined directly using the machine itself. The manufacturing process has therefore been designed to be progressive: the machine gradually produces the parts required for its own completion.



Machine frame under assembly

Precision Engineering – 5-Axis CNC Milling Machine

CONCLUSION :

This project has enabled me to design a complete machine tool while integrating the constraints of rigidity, precision, motorization, and control from the very first stages. From the design of the mechanical architecture to the construction of the frame and the integration of the electronics, it has led me to address industrial design as a whole.

Still under construction, this project also embodies a progressive approach in which the machine itself is used to manufacture some of the components required for its own completion. Above all, it has allowed me to develop a comprehensive approach to mechanical design by continuously confronting CAD design decisions with the practical constraints of manufacturing and assembly.

Key Figures:

- 5 axes
- 2.2 kW spindle power
- NEMA 34 closed-loop motor system
- 100 kg machine weight once completed

The development of this machine has enabled me to further strengthen my skills in mechanical design and manufacturing, while also developing a more comprehensive understanding of mechatronic system integration. The project is still ongoing and remains a long-term development effort that I continue

Skills Developed:

Design

- Design of a complete machine tool
- Sizing and integration of mechanical components

Manufacturing

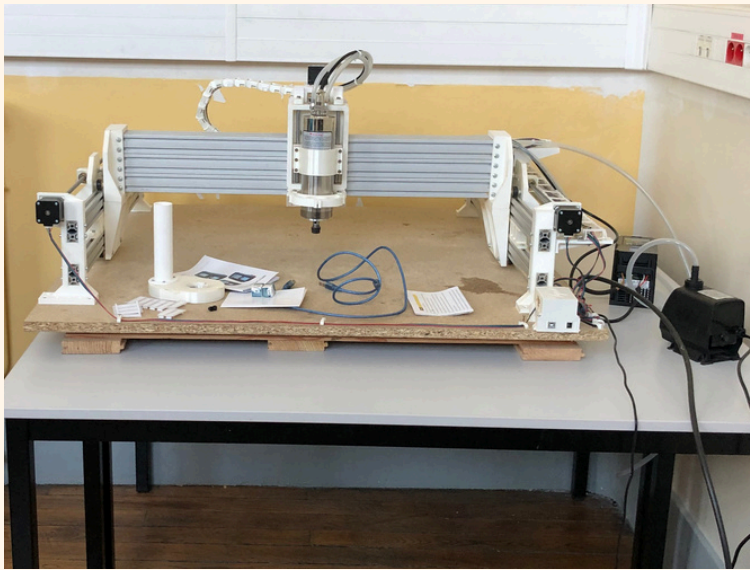
- CNC machining
- Assembly and geometric alignment

Mechatronics

- 5-axis control
- Integration of motors, drivers, and sensors
- Arduino Mega and GRBL Mega 5X

Other projects

The projects presented earlier represent a selection of the work that best defines my professional journey. Other projects—whether one-off assignments or initiatives in different fields—complement this experience and demonstrate the diversity of techniques I have employed.



3-axis CNC milling machine:
Construction of a 3-axis CNC wood milling machine based on plans.



Wood-carbon foils:
Design and manufacture of wood and carbon fiber foils for a radio-controlled sailboat.



Modular wind tunnel:
Resizing and 3D-printed fabrication of an experimental wind tunnel. Design of a 2-axis test rig to measure lift and drag.